# Licensing & TCO Deep-Dive

## Broadcom Changes, True Cost Modeling & 5-Year Savings

Broadcom's acquisition of VMware changed everything: perpetual licenses eliminated, prices increased 2-5x, bundles forced, and small customers dropped. This presentation provides line-by-line cost analysis comparing VMware's new pricing to KVM alternatives at every scale — from 10 VMs to 1,000 VMs — with 5-year projections.

Broadcom Impact | Per-Core Pricing | Bundle Analysis | 5-Year TCO | Hidden Costs | Migration ROI

# Broadcom Licensing Changes

What changed in 2024-2025 and why organizations are forced to re-evaluate.

## Key Changes Under Broadcom

| Change | Before (VMware) | After (Broadcom) | Impact |
|---|---|---|---|
| License model | Perpetual + annual SnS | Subscription only (no perpetual) | Never own software; pay forever or lose access |
| Pricing unit | Per-socket | Per-core (min 16 cores) | 2-5x price increase for modern CPUs (32-128 cores) |
| Product bundles | A la carte (buy what you need) | Forced bundles (VMware Cloud Foundation) | Pay for NSX, vSAN even if not used |
| Small customer support | Partners, resellers, online | Direct Broadcom only (many partners dropped) | 10-50 VM customers lose access to support |
| Perpetual license status | Valid indefinitely | End-of-support; no patches after renewal lapse | Existing perpetual licenses become worthless |
| Multi-year discounts | Available (10-20%) | 3-year minimum commitment required | Locked in for 3 years at inflated prices |
| Free ESXi | Free hypervisor available | Discontinued | Lab/test/small deployments need paid license |
| Audit frequency | Occasional | Aggressive compliance audits | Surprise true-up costs; audit disruption |

### Price Increase Examples (Real Customer Reports)

A 100-VM customer on vSphere Standard saw their renewal quote increase from $45K/yr to $187K/yr (315% increase). A 500-VM enterprise was quoted $1.2M/yr for VMware Cloud Foundation vs $350K/yr previously. Small businesses with 10-20 VMs received "minimum order" requirements of $100K+ or were told to find a different vendor.

# Per-Core Pricing Analysis

How per-core pricing dramatically increases costs for modern hardware.

## Per-Socket vs Per-Core Cost Impact

| Server Config | Sockets | Cores/Socket | Total Cores | Old Price (per-socket) | New Price (per-core) | Increase | Notes |
|---|---|---|---|---|---|---|---|
| Entry (old HW) | 1 | 8 | 8 | $2,875 | $4,600 (16 min) | **60%** | 16-core minimum applies |
| Small server | 1 | 16 | 16 | $2,875 | $4,600 | **60%** | Matches minimum |
| Mid-range | 2 | 24 | 48 | $5,750 | $13,800 | **140%** | Common config today |
| Standard | 2 | 32 | 64 | $5,750 | $18,400 | **220%** | Dell R760 typical |
| High-density | 2 | 48 | 96 | $5,750 | $27,600 | **380%** | AMD EPYC |
| Maximum | 2 | 64 | 128 | $5,750 | $36,800 | **540%** | AMD EPYC 9654 |

Based on VMware vSphere Standard per-core pricing (~$287.50/core). Enterprise Plus is higher. VCF bundle adds NSX + vSAN costs.

## The Modern CPU Penalty

### Why Per-Core Hits Hardest Now

- CPU cores doubled every 3-4 years since 2018
- 2018: typical server = 2x 12-core = 24 cores
- 2024: typical server = 2x 32-core = 64 cores
- 2025: AMD EPYC = 2x 96-core = 192 cores
- Per-core pricing punishes hardware upgrades
- Perverse incentive: buy fewer cores (slower) to save license cost
- Or: switch to KVM (license cost = $0 regardless of core count)

### KVM: $0 Per Core, Forever

- KVM is part of the Linux kernel (GPLv2)
- No per-core, per-socket, per-VM, or per-CPU licensing
- Buy the biggest CPU you want — no penalty
- AMD EPYC 9654 (192 cores): VMware = $55,200/yr; KVM = $0
- Hardware upgrades never increase software cost
- No audit risk, no true-up, no surprise bills
- Optional: RHEL subscription ($799/host/yr) for support

# 5-Year TCO Comparison

Complete total cost of ownership for three deployment sizes.

## Scenario A: Small (20 VMs, 2 Hosts, 32 cores each)

| Cost Category | VMware (5yr) | KVM + RHEL (5yr) | KVM + Rocky (5yr) | Notes |
|---|---|---|---|---|
| Hypervisor licensing | $92,000 | $0 | $0 | VCF per-core x 64 cores x 5yr |
| OS subscriptions | Included | $8,000 | $0 | RHEL: $799/host/yr x 2 x 5 |
| Hardware (2 servers) | $24,000 | $24,000 | $24,000 | Same hardware |
| Migration cost | $0 | $10,000 | $10,000 | One-time labor (h2kvmctl included*) |
| Training | $0 | $5,000 | $5,000 | KVM training for 2 admins |
| Operations (5yr) | $125,000 | $125,000 | $125,000 | Same admin effort |
| **5-Year Total** | **$241,000** | **$172,000** | **$164,000** | |
| **5-Year Savings** | — | **$69,000 (29%)** | **$77,000 (32%)** | |

## Scenario B: Medium (100 VMs, 5 Hosts, 64 cores each)

| Cost Category | VMware (5yr) | KVM + RHEL (5yr) | KVM + Rocky (5yr) | Notes |
|---|---|---|---|---|
| Hypervisor licensing | $460,000 | $0 | $0 | VCF x 320 cores x 5yr |
| OS subscriptions | Included | $20,000 | $0 | RHEL x 5 hosts x 5yr |
| Hardware (5 servers) | $125,000 | $125,000 | $125,000 | Same |
| Migration | $0 | $45,000 | $45,000 | One-time |
| Training | $0 | $15,000 | $15,000 | 3 admins |
| Operations (5yr) | $375,000 | $375,000 | $375,000 | Same |
| **5-Year Total** | **$960,000** | **$580,000** | **$560,000** | |

| | | | |
|---|---|---|---|
| **5-Year Savings** | — | **$380,000 (40%)** | **$400,000 (42%)** |

# Scenario C: Large Enterprise

500 VMs, 20 hosts, 96-core AMD EPYC — where per-core pricing hurts most.

## 500 VMs, 20 Hosts, 96 Cores Each (1,920 total cores)

| Cost Category | VMware VCF (5yr) | KVM + RHEL (5yr) | KVM + Rocky (5yr) | Notes |
|---|---|---|---|---|
| Hypervisor licensing | $2,760,000 | $0 | $0 | VCF ~$287/core x 1920 x 5yr |
| vCenter + add-ons | $175,000 | $0 | $0 | vCenter + vROps + Log Insight |
| OS subscriptions | Included | $80,000 | $0 | RHEL x 20 hosts x 5yr |
| Hardware (20 servers) | $800,000 | $800,000 | $800,000 | Same hardware |
| Migration | $0 | $90,000 | $90,000 | One-time (500 VMs, 2 engineers, 12 weeks) |
| Training | $0 | $25,000 | $25,000 | 5 admins, certifications |
| Operations (5yr) | $1,250,000 | $1,250,000 | $1,250,000 | Same staffing |
| **5-Year Total** | **$4,985,000** | **$2,245,000** | **$2,165,000** | |
| **5-Year Savings** | — | **$2,740,000 (55%)** | **$2,820,000 (57%)** | |

## Summary Across All Scales

### 5-Year Savings by Scale

| Scale | VMs | VMware 5yr | KVM 5yr | Savings |
|---|---|---|---|---|
| Small | 20 | $241K | $164K | **$77K (32%)** |
| Medium | 100 | $960K | $560K | **$400K (42%)** |
| Large | 500 | $4,985K | $2,165K | **$2,820K (57%)** |

**Larger scale = larger savings %** (per-core licensing scales linearly, KVM stays at $0)

### Hidden VMware Costs Not Included

- Audit true-up penalties (surprise $50K-$200K)
- Price increases at renewal (10-30% typical)
- Consultant fees for VMware-specific projects
- Vendor lock-in exit costs (if delayed)
- Opportunity cost of 3-year lock-in commitment
- Management overhead of license compliance
- **True savings likely 10-20% higher than shown**

### KVM Saves $77K to $2.8M Over 5 Years — Depending on Scale

The math is clear: VMware's per-core subscription model creates an ever-growing cost that scales with hardware upgrades. KVM's cost is $0 regardless of cores, sockets, or VMs. The migration investment (one-time) pays for itself in 3-10 months. Over 5 years, even the smallest deployment saves $77K. hyper2kvm makes the migration technically straightforward — the only remaining question is when, not if.